

Label-Flip Attack Detection via Trust-Weighted Aggregation in Federated Learning for Underground Mine Security

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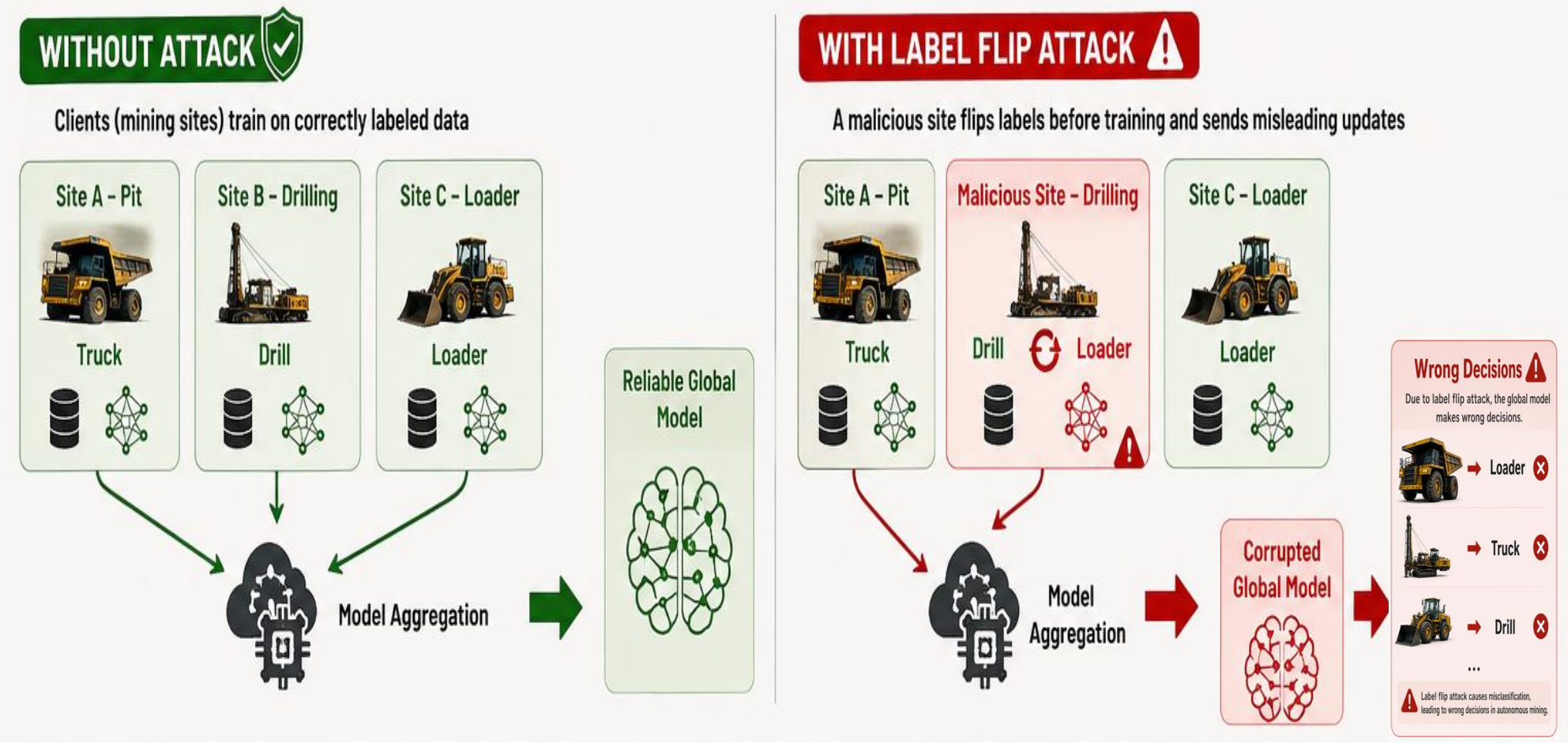
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paper & code

Motivation



Objectives

1. To develop a robust defense framework to protect Federated Learning models from label-flip attacks in underground mining operations, ensuring model integrity and security.
2. Constructing a real-world dataset from experimental mine to mimic Label Flip attack.

Challenges:

1. Label Flip attacks are simple to execute yet catastrophic in underground mines .
2. Non-IID data heterogeneity across mine sites produces naturally diverse gradient patterns, making it extremely difficult to distinguish malicious updates from legitimate local distribution shifts, rendering standard defenses unreliable.

MineSigns Dataset

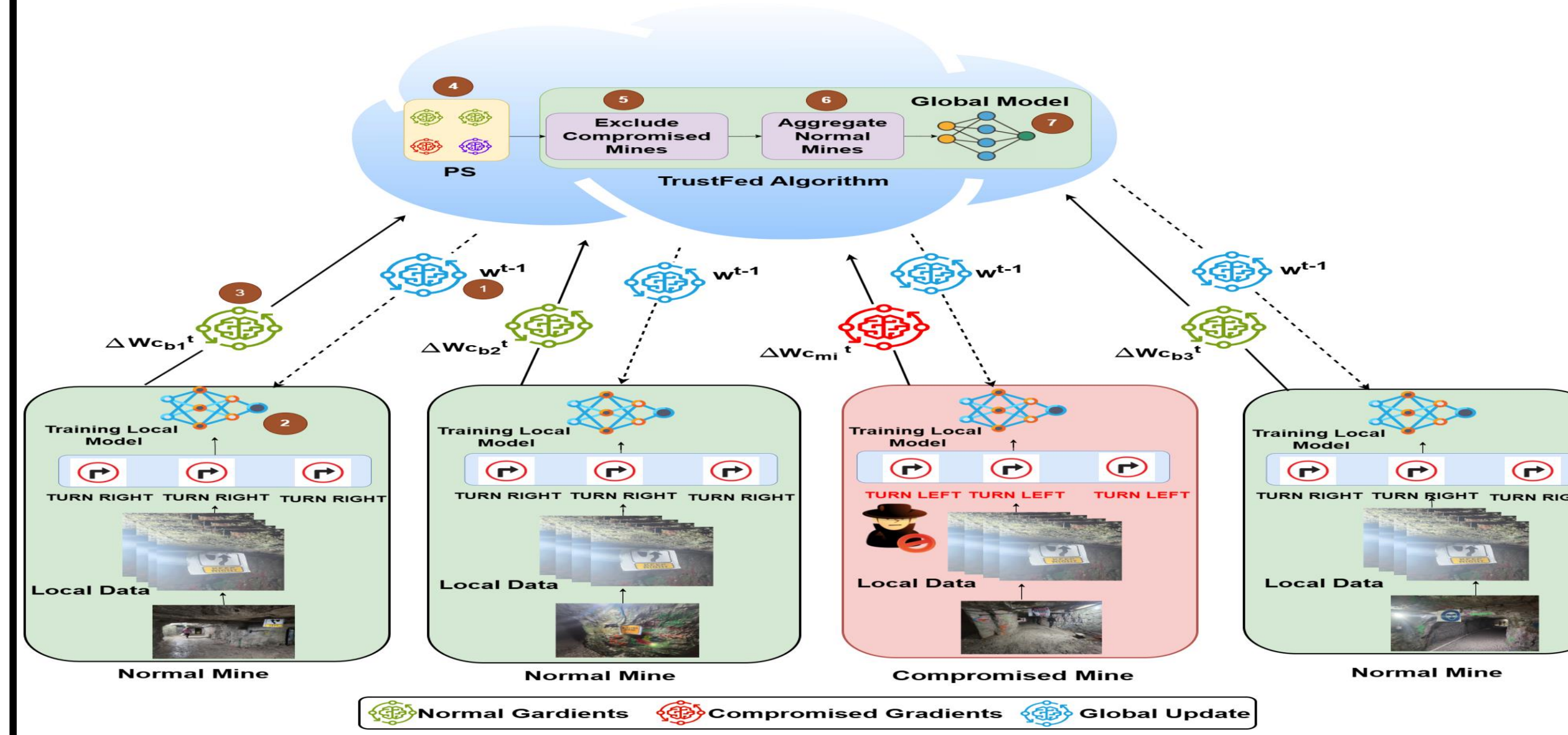
1. Contains 13 important underground mine safety signs .
2. Collected from an experimental mine at Missouri S&T.
3. Supports both IID and non-IID federated learning experiments.
4. Includes realistic variations in illumination, scale, angle, and background conditions.
5. Enables evaluation under adversarial settings such as Label Flip attacks.
6. Helps simulate distributed mine environments with heterogeneous client data.



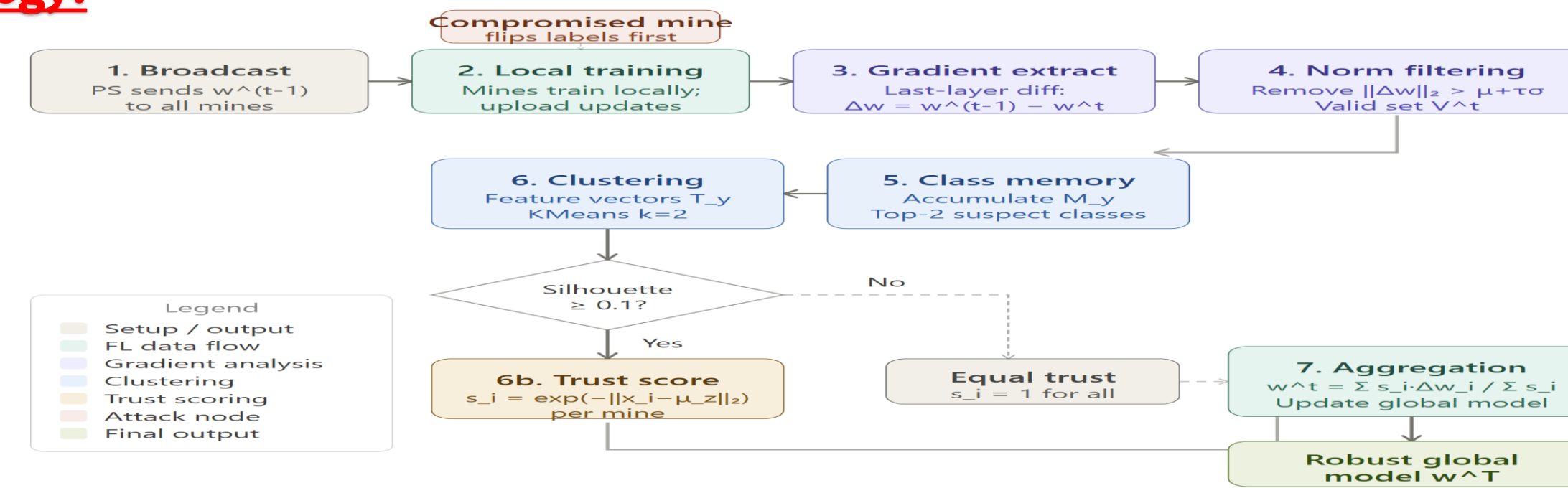
Threat Model

1. Central Server is honest.
2. Central server is solely responsible for identifying and excluding attackers.
3. Total number of malicious mines is lower than total number of normal mines.
4. Before the process begins, a specific set of mines can launch Label Flip attack.
5. No mine can access the model of other or collude with the central server.

TrustFed Framework



Methodology:



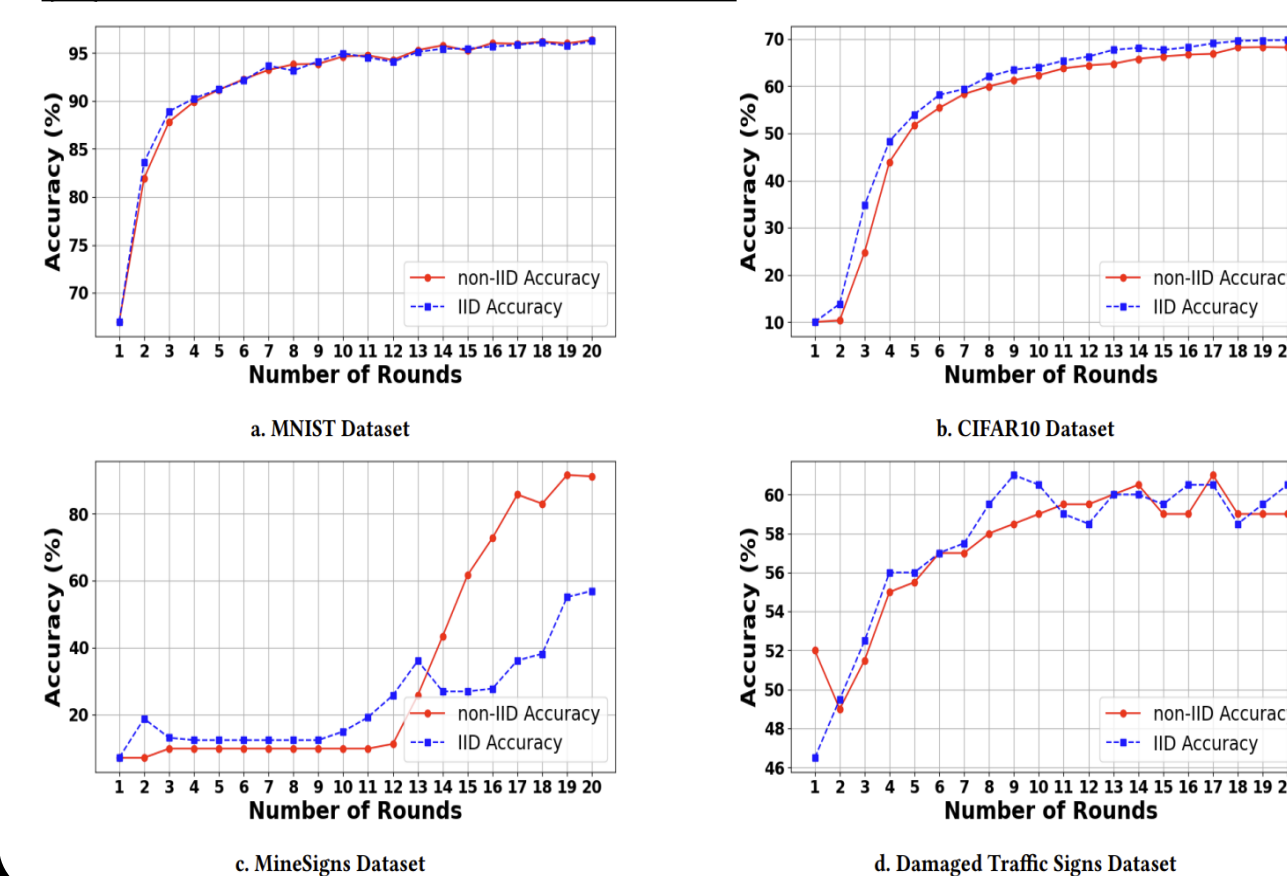
Experiments

Experimental Setup:

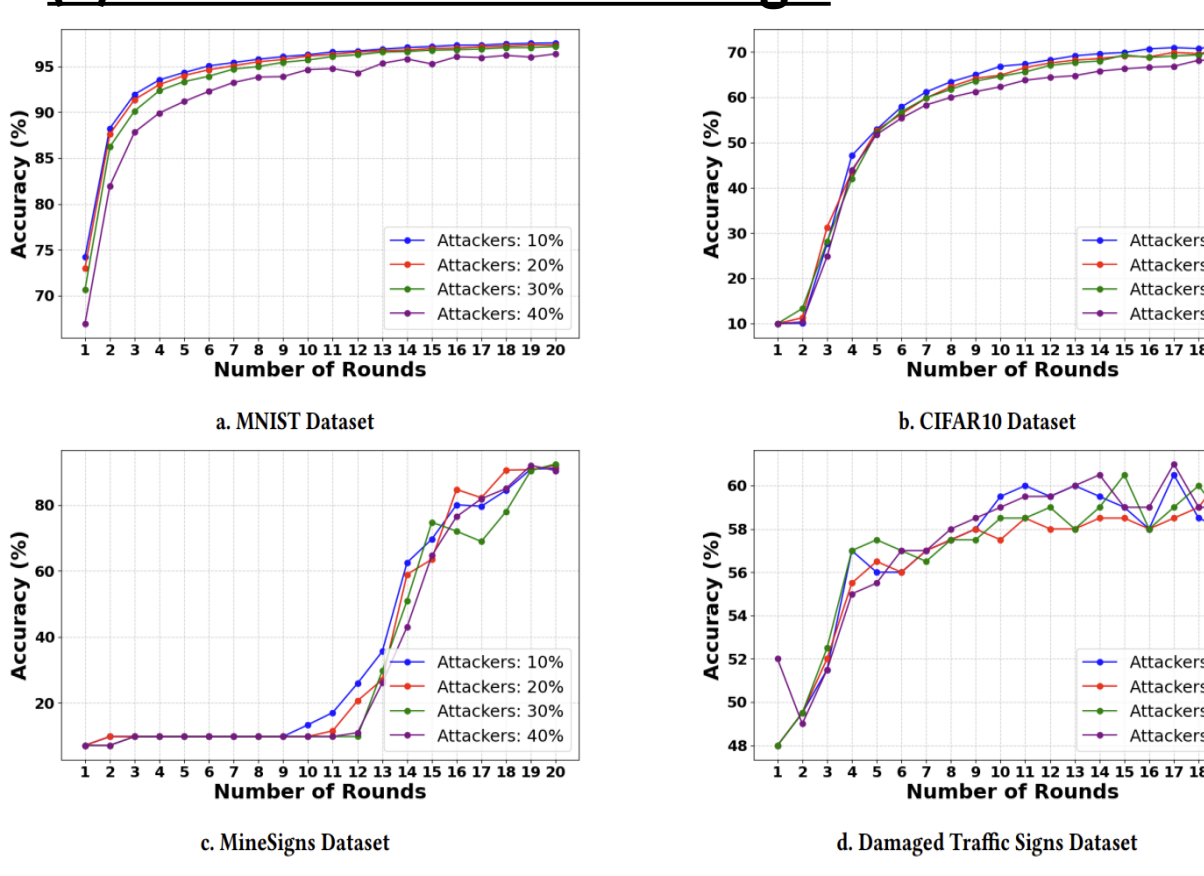
- We evaluate the TrustFed framework on MNIST, Fashion-MNIST, CIFAR10, CIFAR100.
- Application specific dataset- MineSigns[1], Damaged Traffic Signs[2].
- We evaluate the performance using Accuracy, Source class Accuracy, Confusion Matrix and Average Aggregation Time.
- We compare with Mkrum[3], FLAME[4], Lfighter[5], Tolp[6], Tmean[7].

Experimental Results:

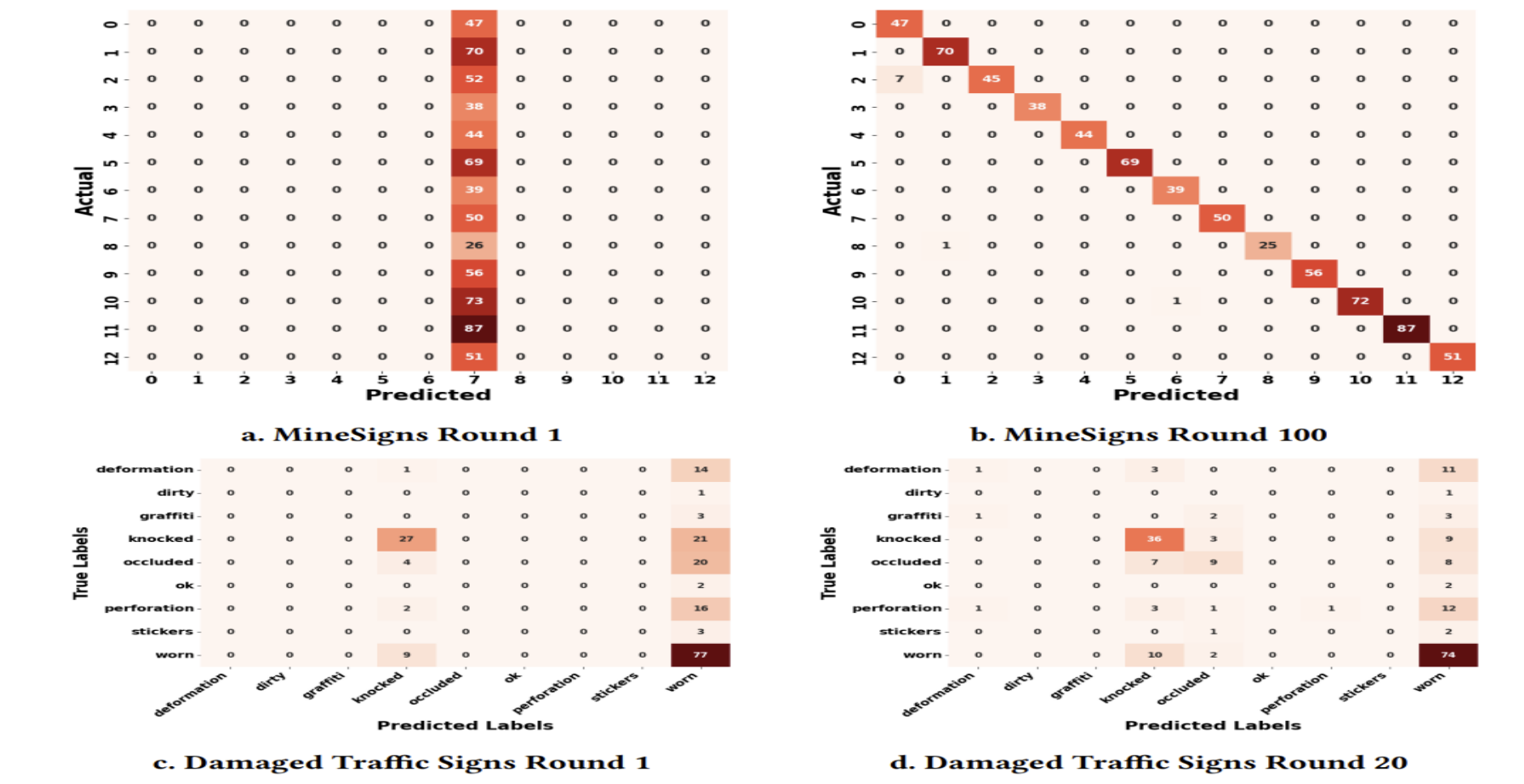
(1) Effect of Data Distribution



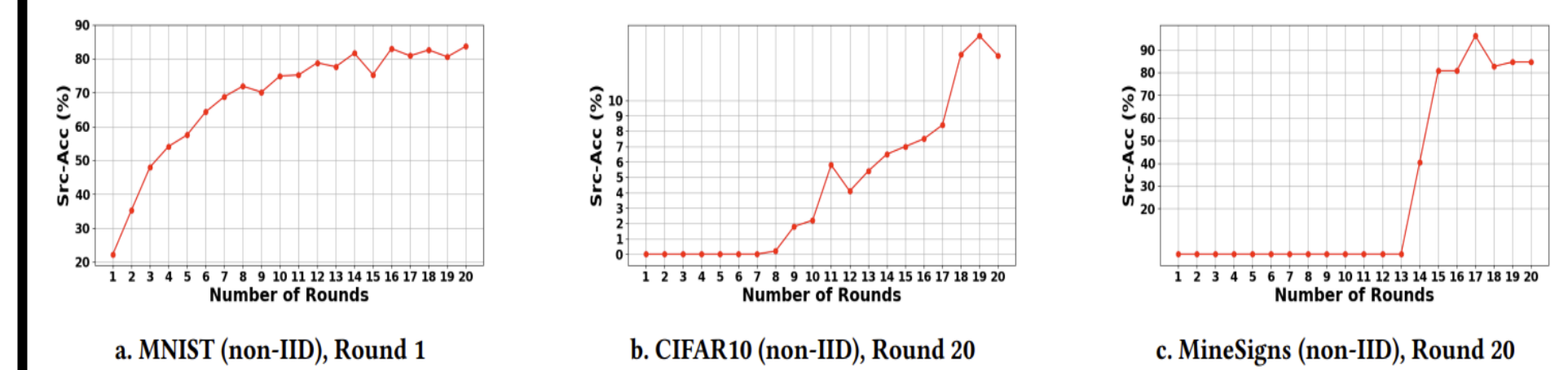
(2) Effect of Attacker Percentage:



(3) Confusion Matrix:



(4) Source-class Accuracy:



(5) Comparison of Average Aggregation Time:

Rounds	FLAME[27]	MKrum[3]	LFighter[16]	Tolp[34]	TMean[36]	TrustFed (Ours)
20	0.073	0.031	0.132	0.271	0.046	0.026
40	0.072	0.036	0.097	0.055	0.054	0.034
60	0.090	0.025	0.074	0.177	0.040	0.026
80	0.070	0.130	0.062	0.263	0.034	0.032
100	0.061	0.092	0.063	0.062	0.035	0.034

Conclusion and Future Work

- Proposed TrustFed, a lightweight and fully unsupervised federated learning defense framework for safety-critical underground mining environments
- Introduces adaptive trust-weighted aggregation to suppress poisoned updates.
- Demonstrated strong robustness against label-flipping attacks across MNIST, CIFAR-10, CIFAR-100, Damaged Traffic Signs, and the proposed MineSigns dataset
- Introduced MineSigns, a dedicated benchmark dataset for robust federated learning research in autonomous mining systems
- Integrate privacy-preserving secure aggregation techniques .
- Extend TrustFed to defend against more advanced attacks including backdoor and adaptive adversarial attacks.
- Expand the MineSigns dataset with additional classes and larger-scale real-world samples
- Explore real-time deployment in fully autonomous and event-driven underground mining ecosystem

References

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- [4] Thien Duc Nguyen, Philip Rieger, Huili Chen, Hossein Yalame, Helen Möllering, Hossein Fereidooni, Samuel Marchal, Markus Miettenen, Azalia Mirhoseini, Shaza Zeinouni, et al. 2022. (FLAME): Taming backdoors in federated learning. In 31st USENIX security symposium (USENIX Security 22), 1415–1432.
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